(TR-104) PYTHON WITH AI/ML

Training Day 53 Report

Date: 25 March 2026

Topic Covered: Smart Task Agent & Memory Management System

The fifty-third day of training focused on revising Transformer concepts and developing a Smart Task Agent with session-based memory management for execution-focused AI workflows.

- Revised Transformer architecture and core concepts related to fine-tuning workflows.
- Revised the process of fine-tuning Transformer models for downstream NLP tasks.
- Implemented agent.py using the Gemini SDK for execution-mode content generation.
- Configured system prompts to generate fully ready-to-post outputs automatically.
- Integrated session-based memory handling using memory.py to maintain conversation history and task tracking.
- Enabled CLI-based commands for viewing history, clearing sessions, and exiting the application.
- Developed memory.py module for persisting conversation turns and task logs.
- Implemented methods for retrieving session history and generating context summaries.
- Added functionalities to list tasks, clear sessions, and summarize previously completed tasks.
- Worked with JSON-based persistence for storing conversational memory and task data.
- Used pathlib for structured file handling and memory storage operations.
- Strengthened understanding of prompt engineering for execution-focused AI agents.
- Improved understanding of session-based memory management in conversational AI systems.

- **GitHub Link:** https://github.com/JaspinderKaurWalia26/agent
- **Blog Link:** https://medium.com/@jaspinderkaurwalia855/building-a-smart-task-agent-using-google-gemini-5048461e356e